Network Traffic Probability Assignment

Applied Statistics | Joint marginal and conditional distributions | 100 points

Purpose

Use network traffic records to construct and interpret joint, marginal, and conditional distributions. Investigate destination ports, firewall outcomes, recorded flow sizes, and average bytes per packet.

Dataset and tools

Download the UCI Internet Firewall Data. Click Download, extract the ZIP file, and open log2.csv. The dataset contains 65,532 records from a university firewall.

https://archive.ics.uci.edu/dataset/542/internet+firewall+data

You may use Python, Include your code or a clear description of your code and procedures.

Treat each row as one traffic record. Your probabilities describe a record selected uniformly at random from the dataset.

Part 1 Prepare the data 10 points

Use Destination Port and Action to define two categorical random variables.

Variable	Category	Definition
X Destination port	Port 53	Destination Port = 53
X Destination port	Port 80	Destination Port = 80
X Destination port	Port 443	Destination Port = 443
X Destination port	Other	Any other destination port
Y Firewall outcome	Allowed	Action = allow
Y Firewall outcome	Not allowed	Any other recorded Action value

Port numbers are grouping labels; a port number alone does not prove which application generated the traffic.

1. Report the original number of records and list the distinct Action values.
2. Check the two required columns for missing or invalid values. Exclude affected records and report how many were excluded.
3. Report the final number of usable records, N. Retain repeated rows because identical recorded values do not necessarily represent duplicate events. Use this same cleaned dataset throughout.

Part 2 Joint distribution 20 points

A. Create the following frequency table from the cleaned dataset.

Destination-port category	Allowed	Not allowed	Total
Port 53			
Port 80			
Port 443			
Other			
Total			N

B. Construct the joint probability table by dividing each interior cell count by N.

$$\hat{P}(X = x, Y = y) = \frac{n_{xy}}{N}$$

Here n_{xy} is the number of records in cell (x, y). The hat indicates a probability estimated from the observed data.

1. Verify that the eight joint probabilities sum to 1.
2. Find the probability that a randomly selected record uses port 443 and is allowed. (What percentage of all records use port 443 and are allowed?)
3. Find the probability that a record uses port 53 and is not allowed. (What percentage of all records use port 53 and are not allowed?
4. Find the probability that a record uses either port 80 or port 443, and is allowed. (What percentage of all records use either port 80 or port 443 and are allowed?)
5. Interpret your result for port 443 and Allowed in a complete sentence.

Part 3 Marginal distributions 10 points

Calculate both marginal distributions by summing the joint probabilities.

$$\hat{P}(X = x) = \sum_y \hat{P}(X = x, Y = y)$$

$$\hat{P}(Y = y) = \sum_x \hat{P}(X = x, Y = y)$$

1. Present a marginal probability table for each variable and verify that each distribution sums to 1.
2. Identify the most common destination-port category.
3. Report the overall probability that a record is allowed.
4. Explain what information is lost when examining only the marginal distributions.

Part 4 Conditional distributions 20 points

Calculate the following probabilities. For each, show the numerator and denominator, report the probability, and interpret it in context.

$$P(Y = Allowed \mid X = Port\ 443),\ \hat{P}(X = Port\ 443 | Y = Allowed\ )$$

$$P(Y = Not\ allowed \mid X = Port\ 53), \hat{P}(\ X = Port\ 53 \mid Y = Not\ allowed)$$

If the conditioning category has no records, report the probability as undefined and explain why.

1. Construct the conditional distribution of firewall outcome within each destination-port category. Verify that each nonempty row sums to 1.
2. Create a bar chart comparing the probability of being allowed across the four port categories. Provide the underlying values in a table.
3. Explain why P(Allowed | Port 443) and P(Port 443 | Allowed) answer different questions.

Part 5 Interpretation and independence 15 points

1. If X and Y were independent, they would satisfy the relationship below. Compare the observed joint probability with the product of the marginal probabilities for two cells in your table. Show the calculations.

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

2. Compare the overall allowance probability with the allowance probability within each port category. Does the dataset suggest an association between port category and firewall outcome? Support your answer with numbers. A formal hypothesis test is not required.
3. Explain why “not allowed” does not automatically mean malicious traffic.
4. Write a 100–150-word conclusion summarizing your findings and explaining why results from this firewall may not represent all networks.

Part 6 Flow size and average packet size 25 points

The dataset also contains Bytes and Packets. Treat recorded flow size as the total Bytes in one traffic record. Calculate average bytes per packet by dividing Bytes by Packets for that same record.

$$Average\ bytes\ per\ packet\ = \frac{Bytes}{Packets}$$

This ratio is a record-level average, not an individual packet length. The data do not contain individual packet sizes or the variation of packet sizes within a record. Use "average bytes per packet" when labeling your results.

For this part, use records with nonmissing numeric Bytes ≥ 0 and a positive integer Packets value, as well as valid Destination Port and Action. Keep legitimate zero-byte records. Report the excluded count and the resulting sample size M. Use M for every probability in this part; retain N for Parts 1–5.

Define two additional categorical variables using these assignment-specific cutoffs. These are analysis choices, not universal network standards.

Variable	Small	Medium	Large
F Recorded flow size	0 ≤ Bytes < 1,000	1,000 ≤ Bytes < 10,000	Bytes ≥ 10,000
A Average bytes per packet	0 ≤ average < 200	200 ≤ average < 800	average ≥ 800

1. Construct a 3 × 3 frequency table of F versus A, including row and column totals. Convert it to a joint probability table by dividing each interior count by M. Verify that the nine joint probabilities sum to 1. (6 points)
2. Calculate both marginal distributions. Report the probabilities of a Large flow and a Large average bytes per packet. Interpret each result. (4 points)
3. Calculate P(F = Large, A = Small), P(A = Large | F = Large), and P(F = Large | A = Large). Show the counts and denominator for each. If a conditioning category is empty, report the conditional probability as undefined. (6 points)
4. Construct a joint table of F versus firewall outcome Y. Calculate P(Y = Allowed | F = Large) and compare it with the overall allowance probability in these M records. (5 points)
5. Explain how a flow can have many bytes but a small average bytes per packet. Describe how changing the category cutoffs could change your probability tables. Explain why this analysis describes randomly selected records rather than randomly selected individual packets. (4 points)

Submission requirements

Submit a report containing your answers, calculations, tables, and chart; your code, workbook, or reproducible software steps; and a citation or link to the dataset.

Report probabilities to four decimal places, but use unrounded values in intermediate calculations. Include one worked example of each probability type showing the underlying counts and denominator.

Optional extension: Use Bayes' theorem to calculate P(F = Large | A = Large), and verify that it matches the proportion calculated directly from the M records. You may also explore the Packets or Elapsed Time (sec) variables with your own clearly defined categories.

Dataset citation: Internet Firewall Data (2019). UCI Machine Learning Repository. https://doi.org/10.24432/C5131M. Licensed under CC BY 4.0.